Intermediate Elective Option 2

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```r
# general use
library(tidyverse)

# file organization
library(here)

# color display helper
library(prismatic)

# additional scales and formatting
library(scales)

# font support
library(showtext)

# add Bell MT equivalent from Google Fonts
font_add_google("Playfair Display", "playfair")
showtext_auto()
```

```r
# read in vegetation transect data
veg <- read_csv(here("data", "veg.csv"))

# read in pool-level metadata
metadata <- read_csv(here("data", "vp_veg_metadata.csv"))
```

1. Explain your inspiration

I chose Tobias Stalder's "Hiking Locations in Washington" circular bar chart as my inspiration. This visualization uses `coord_polar()` in `ggplot2` to arrange grouped bars radially. His visualization contains bar height, fill color, and an overlaid point and segment.

This format makes sense for the vernal pool vegetation dataset because the data contains multiple discrete pools (analogous to Stalder's hiking regions), each with several continuous summary metrics worth comparing in a single figure. The circular layout compares many pools at once without a cluttered axis and the circular layout feels fitting for a dataset about pools.

My visualization will display the following variables from `veg.csv` and `vp_veg_metadata.csv`:

- **Vernal pool** (`transect_name`): the categorical x-axis, arranged radially
- **Cumulative percent cover** (`sum_pc`): mapped to bar height
- **Number of plant species recorded** (count of distinct psoc values per pool): mapped to fill color via a continuous gradient
- **Mean native percent cover** (filtered from `cover_category == "NATIVE COVER"`): mapped to the point and segment overlaid on each bar

2. Plan your figure

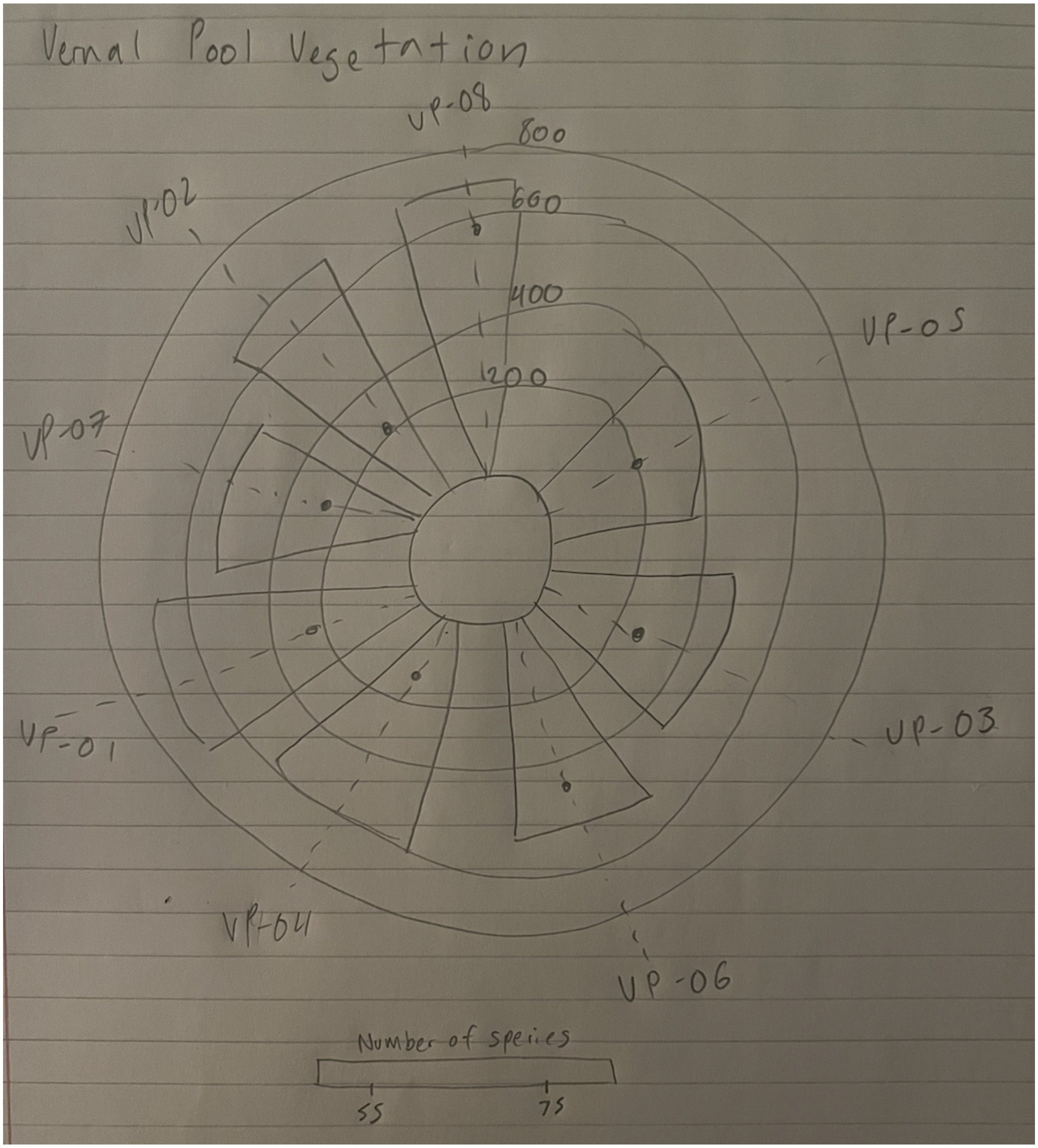


Figure 1: Sketch of vernal pool circular bar chart

3. Code your figure

```r
# create new object from veg
pools_vp <- veg |>
  # filter to only vernal pool transects
  filter(str_detect(transect_name, "VP")) |>
  # complete all combinations of year, pool, species, distance
  # fill missing percent cover values with 0
  complete(year, transect_name, psoc, transect_distance,
           fill = list(percent_cover = 0)) |>
  # group by year, pool, and species
  group_by(year, transect_name, psoc, cover_category) |>
  # calculate total percent cover across all quadrats
  summarize(sum_pc = sum(percent_cover, na.rm = TRUE),
            .groups = "drop") |>
  # create year_pool column from year and transect name, keep originals
  unite("year_pool", year, transect_name, remove = FALSE) |>
  # join metadata to get number of quadrats per pool-year
  left_join(metadata, by = "year_pool") |>
  # calculate mean percent cover from total / number of quadrats
  mutate(mean_pc = sum_pc / num_quad) |>
  # rename transect_name.x back to transect_name after join conflict
  rename(transect_name = transect_name.x) |>
  # drop redundant columns brought in by metadata join
  select(-year.x, -year.y, -transect_name.y)

# calculate cumulative cover and mean native cover per pool
pools_vp |>
  group_by(transect_name) |>
  summarise(
    # sum mean percent cover across all years and species per pool
    sum_cover    = sum(mean_pc, na.rm = TRUE),
    # sum of mean percent cover for native species only
    native_cover = sum(mean_pc[cover_category == "NATIVE COVER"],
                       na.rm = TRUE)
  ) |>
  # round native cover to match Stalder's mean_gain rounding
  # store as summary_stats
  mutate(native_cover = round(native_cover, digits = 0)) -> summary_stats

# count distinct species with non-zero cover per pool
pools_vp |>
```

```r
  group_by(transect_name) |>
  summarise(n = n_distinct(psoc[mean_pc > 0])) -> species_counts

# join species counts back to summary stats
left_join(summary_stats, species_counts, by = "transect_name") -> summary_all
```

```r
# display color palette in console for reference
prismatic::color(c("#6C5B7B", "#C06C84", "#F67280", "#F8B195"))
```

```
<colors>
#6C5B7BFF #C06C84FF #F67280FF #F8B195FF
```

```r
# store figure as object for saving
vp_plot <- ggplot(summary_all) +

  # make custom panel grid
  geom_hline(yintercept = 0,   color = "lightgrey") +
  geom_hline(yintercept = 200, color = "lightgrey") +
  geom_hline(yintercept = 400, color = "lightgrey") +
  geom_hline(yintercept = 600, color = "lightgrey") +
  geom_hline(yintercept = 800, color = "lightgrey") +

  # bars: pool ordered by cumulative cover, filled by species count
  geom_col(aes(
    x    = reorder(str_wrap(transect_name, 5), sum_cover),
    y    = sum_cover,
    fill = n),
    position    = "dodge2",
    show.legend = TRUE,
    alpha       = .9) +

  # fill gradient and legend title for number of species per pool
  scale_fill_gradientn("Number of species",
    colours = c("#6C5B7B", "#C06C84", "#F67280", "#F8B195")) +

  # lollipop dot: mean native cover per pool
  geom_point(aes(
    x = reorder(str_wrap(transect_name, 5), sum_cover),
    y = native_cover),
    size  = 3,
    color = "gray12") +
```

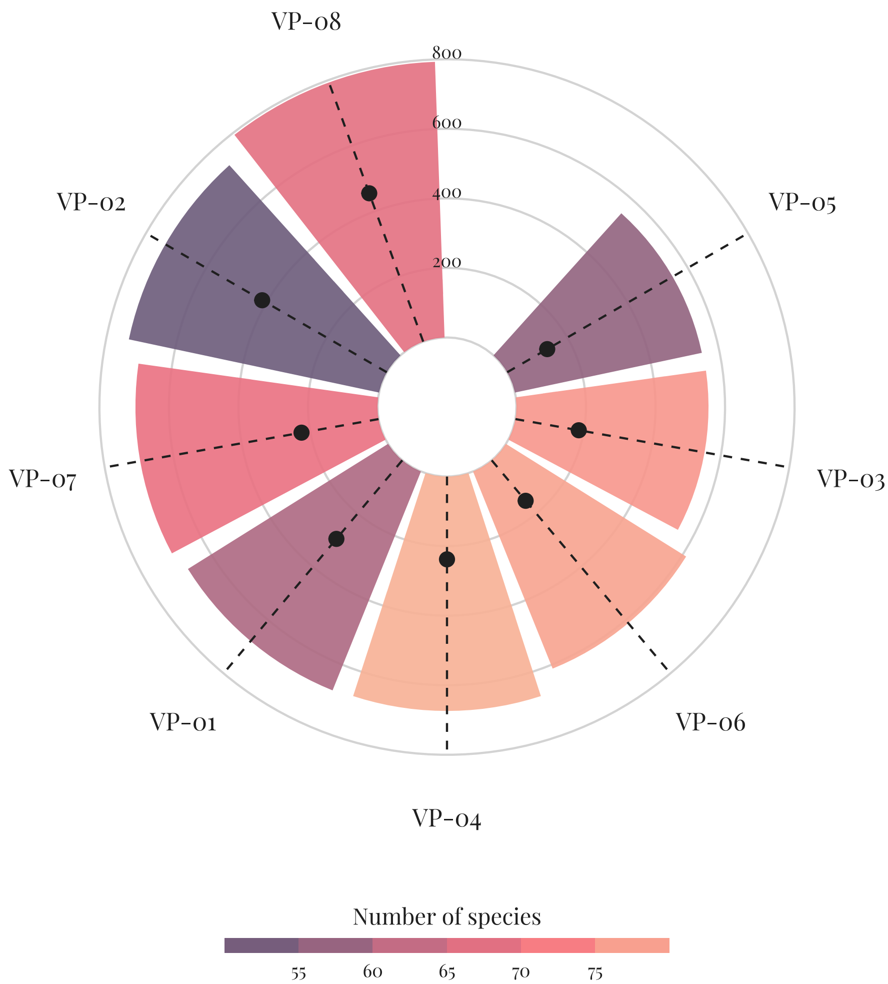
```
# lollipop shaft for native cover per pool
geom_segment(aes(
  x    = reorder(str_wrap(transect_name, 5), sum_cover),
  y    = 0,
  xend = reorder(str_wrap(transect_name, 5), sum_cover),
  yend = 800),
  linetype = "dashed",
  color    = "gray12") +

# annotate bars and lollipops so reader understands scaling
# annotate scale: 200
annotate(x = -0.5, y = 220, label = "200", geom = "text",
       color = "gray12", family = "playfair", size = 3) +
# annotate scale: 400
annotate(x = -0.5, y = 420, label = "400", geom = "text",
       color = "gray12", family = "playfair", size = 3) +
# annotate scale: 600
annotate(x = -0.5, y = 620, label = "600", geom = "text",
       color = "gray12", family = "playfair", size = 3) +
# annotate scale: 800
annotate(x = -0.5, y = 820, label = "800", geom = "text",
       color = "gray12", family = "playfair", size = 3) +

# scale y axis so bars don't start in the center
scale_y_continuous(
  limits = c(-200, 850),
  expand = c(0, 0),
  breaks = c(0, 200, 400, 600, 800)) +

# title, subtitle, caption
labs(
  title    = "\nVernal Pool Vegetation",
  subtitle = paste("\nThis visualization shows the cumulative percent
   ↪  cover",
               "across all species and survey years,",
               "the number of species recorded (bar color),",
               "and the total native cover per pool (dot).\n",
               "VP-08 has the highest cumulative cover,",
               "while VP-03 has the greatest number of species recorded.",
               sep = "\n"),
  caption  = "\nSource: Cheadle Center for Biodiversity and Ecological
   ↪  Restoration | Visualization: Macey Hartmann | Template: Tobias
   ↪  Stalder · github.com/toebR/Tidy-Tuesday",
```

```r
    fill     = "Number of species (color)") +

  # wrap bars into circular polar layout
  coord_polar() +

  # theming
  theme(
    legend.position    = "bottom",
    axis.title         = element_blank(),
    axis.ticks         = element_blank(),
    axis.text.y        = element_blank(),
    axis.text.x        = element_text(color = "gray12", size = 12),
    panel.background   = element_rect(fill = "white", color = "white"),
    panel.grid         = element_blank(),
    panel.grid.major.x = element_blank(),
    text               = element_text(color = "gray12", family = "playfair"),
    plot.title         = element_text(face = "bold", size = 25, hjust =
     ↪  0.05),
    plot.subtitle      = element_text(size = 14, hjust = 0.05),
    plot.caption       = element_text(size = 10, hjust = 0.5)) +

  # format the legend color bar
  guides(fill = guide_colorsteps(
    barwidth       = 15,
    barheight      = 0.5,
    title.position = "top",
    title.hjust    = 0.5))

# display figure
vp_plot
```

Vernal Pool Vegetation

This visualization shows the cumulative percent cover
across all species and survey years,
the number of species recorded (bar color),
and the total native cover per pool (dot).

VP-08 has the highest cumulative cover,
while VP-03 has the greatest number of species recorded.



Center for Biodiversity and Ecological Restoration | Visualization: Macey Hartmann | Template: Tobias Stalder · github.com/toeb

```r
# save final figure as png at 100 dpi for readable text size
ggsave(
  filename = here("outputs", "vp_circular_barchart.png"),
  plot     = vp_plot,
  width    = 10,
  height   = 12,
  units    = "in",
  dpi      = 100)
```

4. Describe your process

To adapt Tobias Stalder's code for my dataset, I first found the similarities between his hiking data and the vernal pool vegetation data and his `region` became my `transect_name`, his `sum_length` became my `sum_cover`, his track count (`n`) became my species count (`n_distinct(psoc)`), and his `mean_gain` became my `native_cover`. I then repurposed his wrangling structure using the class key's `complete()` and `left_join()` approach to handle implicit zeros and pool-level metadata before summarizing to one row per pool.

The main challenge I encountered was a column naming conflict after `left_join()` — both `veg` and the metadata contained `transect_name` and `year`, which produced `.x` and `.y` suffixes that broke subsequent `group_by()` calls. I resolved this by adding `rename()` and `select()` steps to clean up the duplicate columns. I also had to troubleshoot the y-axis limits extensively, as my `sum_cover` values (~550–793) were much larger than expected and required expanding `scale_y_continuous()` to prevent bars from being clipped.

This visualization differs from figures I made in class and for assignments as previous visualizations straightforward x and y axes, while this figure uses `coord_polar()` to wrap bars into a circular layout. It also encodes three variables whereas class figures typically encoded one or two variables. Additionally, this figure required significantly more wrangling and customization, including custom annotations, a Google font, and careful management of y-axis limits and color gradients.